



Institut Universitaire des sciences

- Faculté des sciences et de technologie

✓ TD N°5 Réseau II

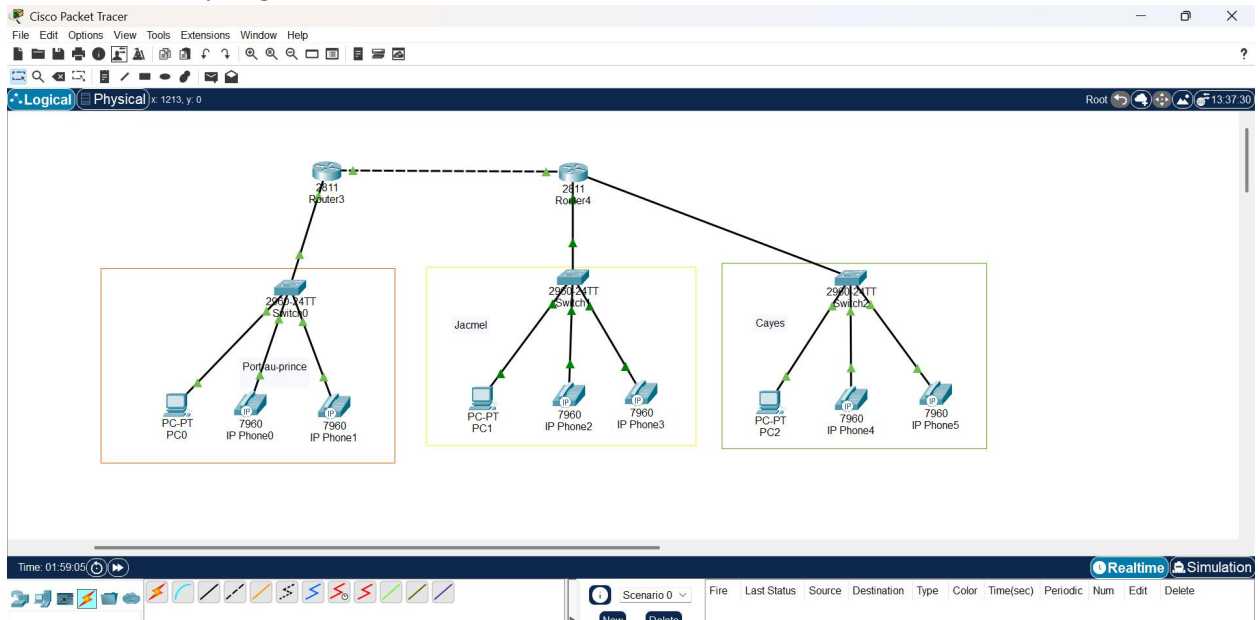
Nom & Prénom : COFFY Cliford

Niveau : L3

Prof. : Ismaël SAINT AMOUR

Date : 26/04/2026

Creation de la topologie





Router3



Physical Config CLI Attributes

MODULES

NM-1E
NM-1E2W
NM-1FE-FX
NM-1FE-TX
NM-1FE2W
NM-2E2W
NM-2FE2W
NM-2W
NM-4A/S
NM-4E
NM-8A/S
NM-8AM
NM-Cover
NM-ESW-161
HWIC-1GE-SFP
HWIC-2T
HWIC-4ESW
HWIC-8A
HWIC-AP-AG-B
WIC-1AM
WIC-1ENET
WIC-1T
WIC-2AM
WIC-2T
WIC-Cover
GLC-LH-SMD

Physical Device View

Zoom In

Original Size

Zoom Out



Customize
Icon in
Physical View



Customize
Icon in
Logical View



The NM-1E features a single Ethernet port that can connect a LAN backbone which can also support either six PRI connections to aggregate ISDN lines, or 24 synchronous/asynchronous ports.

☐ Top



Router3



Physical Config CLI Attributes

MODULES

NM-1E

NM-1E2W

NM-1FE-FX

NM-1FE-TX

NM-1FE2W

NM-2E2W

NM-2FE2W

NM-2W

NM-4A/S

NM-4E

NM-8A/S

NM-8AM

NM-Cover

NM-ESW-161

HWIC-1GE-SFP

HWIC-2T

HWIC-4ESW

HWIC-8A

HWIC-AP-AG-B

WIC-1AM

WIC-1ENET

WIC-1T

WIC-2AM

WIC-2T

WIC-Cover

GLC-LH-SMD

Physical Device View

Zoom In

Original Size

Zoom Out

Customize
Icon in
Physical ViewCustomize
Icon in
Logical View

The NM-2FE2W Module provides two Fast-Ethernet interfaces for use with copper media, in addition to two Wan Interface Card expansion slots. Ideal for a wide range of LAN applications, the Fast Ethernet network modules support many internetworking features and standards.

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Configuration des Routeurs:

Premier Router

 Router3

Physical Config CLI Attributes

IOS Command Line Interface

```
cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory
Processor board ID JAD05190MTZ (4292891495)
2 FastEthernet interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface f0/0
Router(config-if)#ip address 192.168.1.1 255.255.0
      ^
% Invalid input detected at '^' marker.

Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface f0/1
Router(config-if)#ip address 10.10.10.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#
```

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☐ Top

DHCP VoIP

 Router3

Physical Config CLI Attributes

IOS Command Line Interface

```
Router con0 is now available

Press RETURN to get started.

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Router>ip dhcp pool VOIP
^
% Invalid input detected at '^' marker.

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp pool VOIP
Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#option 150 ip 192.168.1.1
Router(dhcp-config)#exit
Router(config)#
```

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CME (Call Manager Express)

 Router3

Physical Config CLI Attributes

IOS Command Line Interface

Router con0 is now available

Press RETURN to get started.

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#telephony-service
Router(config-telephony)#max-ephone 5
Router(config-telephony)#max-dn 5
Router(config-telephony)#ip source-address 192.168.1.1 port 2000
Router(config-telephony)#auto assign 1 to 5
Router(config-telephony)#exit
Router(config)#ephone-dn 1
Router(config-ephone-dn)%%LINK-3-UPDOWN: Interface ephone_dsp DN 1.1, changed state to up
Router(config-ephone-dn)#number 1001
Router(config-ephone-dn)#exit
Router(config)#ephone-dn 2
Router(config-ephone-dn)%%LINK-3-UPDOWN: Interface ephone_dsp DN 2.1, changed state to up
Router(config-ephone-dn)#number 1002
Router(config-ephone-dn)#
```

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Deuxieme Router :

 Router4

Physical Config CLI Attributes

IOS Command Line Interface

If you require further assistance please contact us by sending email to export@cisco.com.

cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory
Processor board ID JAD05190MTZ (4292891495)
2 FastEthernet interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface f0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface f0/1
Router(config-if)#ip address 10.10.10.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#

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Router4

Physical Config CLI Attributes

IOS Command Line Interface

Press RETURN to get started!

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
```

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface f0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface f0/1
Router(config-if)#ip address 10.10.10.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface f0/4
%Invalid interface type and number
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp pool VOIP
Router(dhcp-config)#network 192.168.2.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.2.1
Router(dhcp-config)#option 150 ip 192.168.2.1
Router(dhcp-config)#exit
Router(config)#
```

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CME (Call Manager Express)

 Router4

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#telephony-service
Router(config-telephony)#max-ephone 5
Router(config-telephony)#max-dn 5
Router(config-telephony)#ip source-address 192.168.1.1 port 2000
Router(config-telephony)#auto assign 1 to 5
Router(config-telephony)#exit
Router(config)#ephone-dn 1
Router(config-ephone-dn)%%LINK-3-UPDOWN: Interface ephone_dsp DN 1.1, changed state to up

Router(config-ephone-dn)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#telephony-service
Router(config-telephony)#max-ephone 5
Router(config-telephony)#max-dn 5
Router(config-telephony)#ip source-address 192.168.2.1 port 2000
Router(config-telephony)#auto assign 1 to 5
Router(config-telephony)#exit
Router(config)#ephone-dn 1
Router(config-ephone-dn)#number 2001
Router(config-ephone-dn)#exit
Router(config)#ephone-dn 2
Router(config-ephone-dn)%%LINK-3-UPDOWN: Interface ephone_dsp DN 2.1, changed state to up

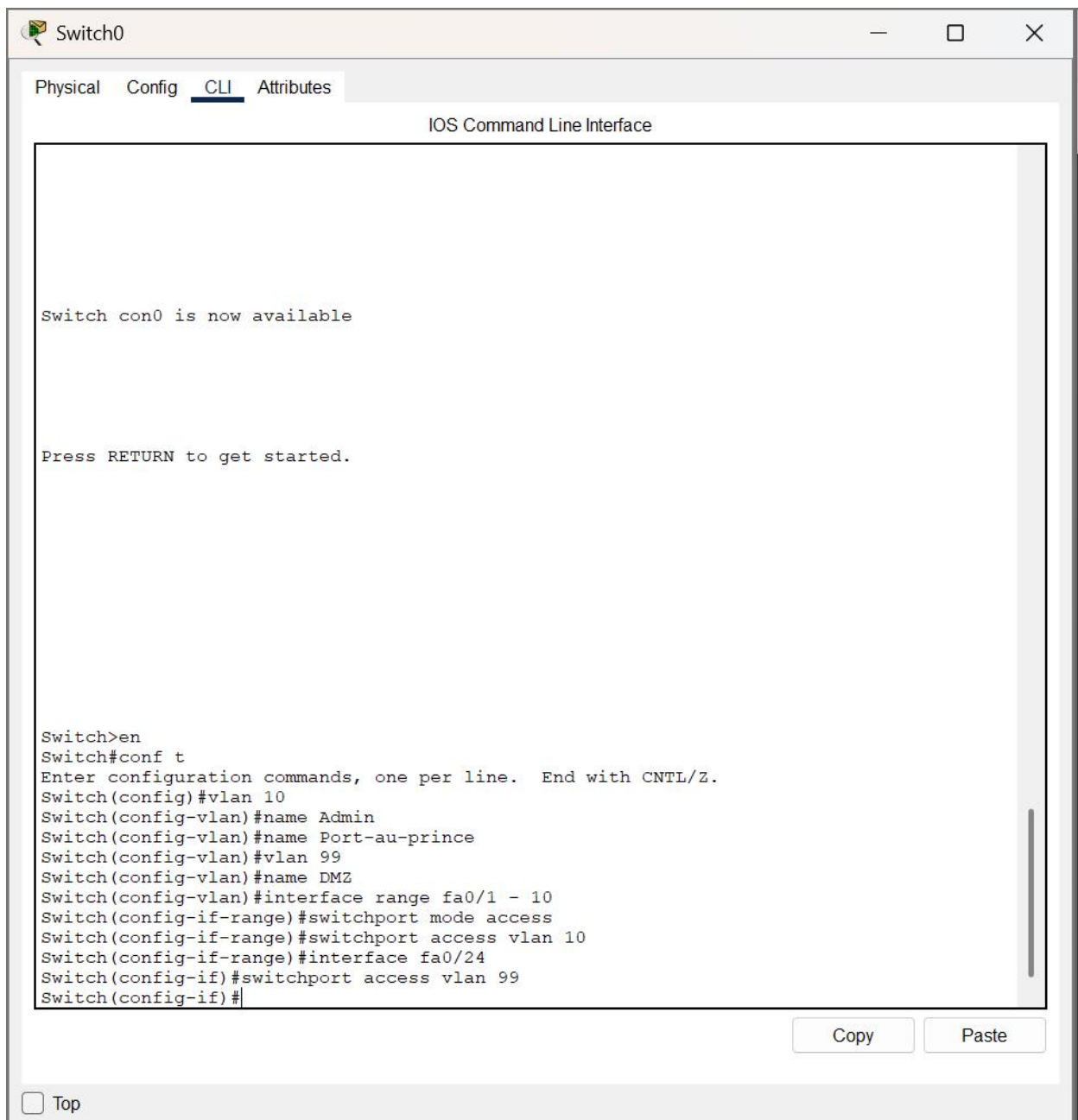
Router(config-ephone-dn)#number 2002
Router(config-ephone-dn)#
```

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Configuration des Switchs:

Switch 1:



Configuration des commutateurs

Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

Switch con0 is now available

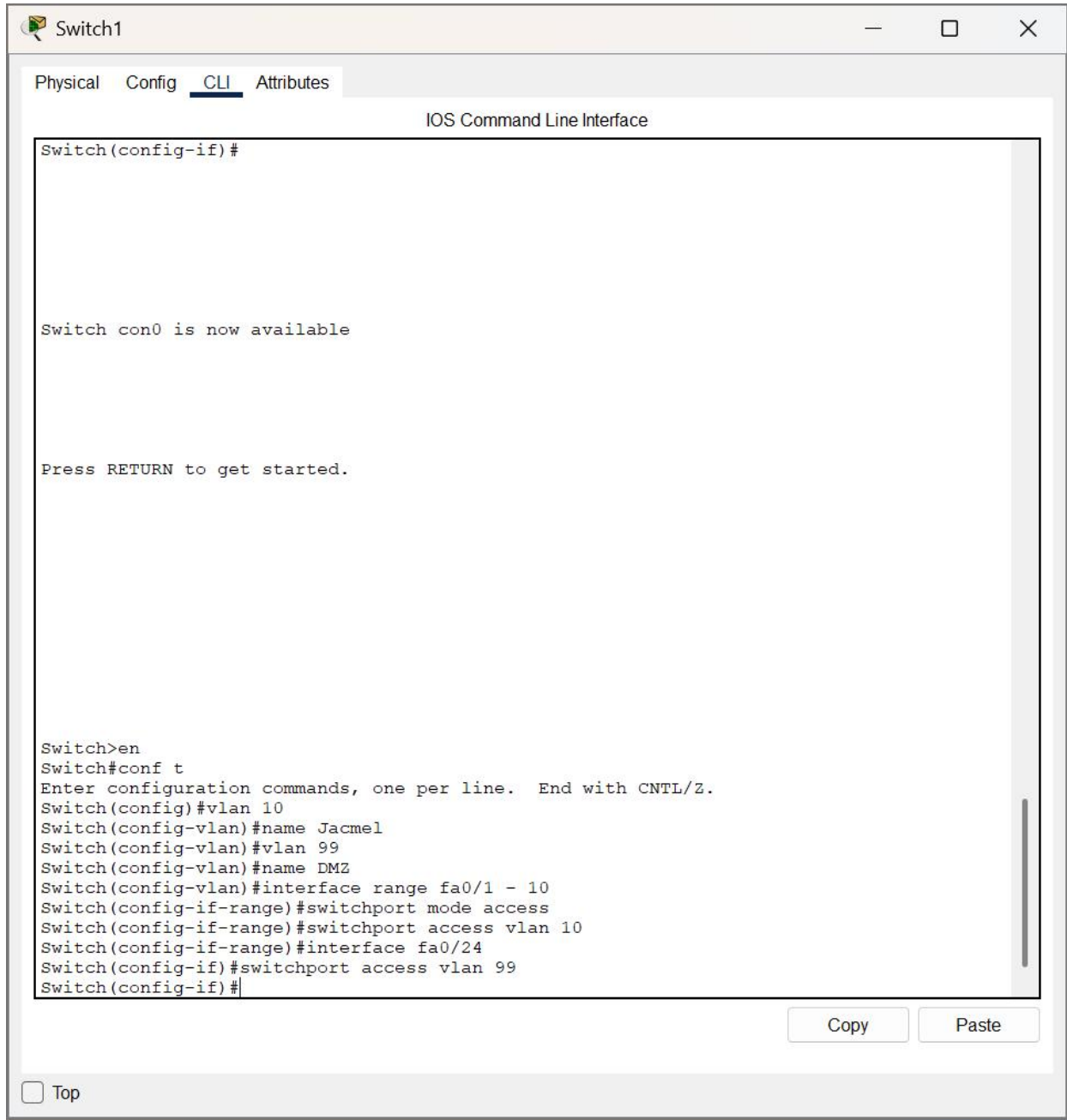
Press RETURN to get started.

Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname S1
S1(config)#interface range fa 0/1-24
S1(config-if-range)#switchport mode access
S1(config-if-range)#switchport voice vlan 1
S1(config-if-range)#exit
S1(config)#
```

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Switch 2:



Switch1

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch(config-if) #  
  
Switch con0 is now available  
  
Press RETURN to get started.  
  
Switch>en  
Switch#conf t  
Enter configuration commands, one per line. End with CNTL/Z.  
Switch(config)#vlan 10  
Switch(config-vlan)#name Jacmel  
Switch(config-vlan)#vlan 99  
Switch(config-vlan)#name DMZ  
Switch(config-vlan)#interface range fa0/1 - 10  
Switch(config-if-range)#switchport mode access  
Switch(config-if-range)#switchport access vlan 10  
Switch(config-if-range)#interface fa0/24  
Switch(config-if)#switchport access vlan 99  
Switch(config-if) #
```

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Configuration des commutateurs

Switch1

Physical Config CLI Attributes

IOS Command Line Interface

Press RETURN to get started.

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name Jacmel
Switch(config-vlan)#vlan 99
Switch(config-vlan)#name DMZ
Switch(config-vlan)#interface range fa0/1 - 10
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#interface fa0/24
Switch(config-if)#switchport access vlan 99
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname S2
S2(config)#interface range fa 0/1-24
S2(config-if-range)#switchport voice vlan 1
S2(config-if-range)#exit
S2(config)#
```

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Switch 3:

 Switch2

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to down
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINK-3-UPDOWN: Interface FastEthernet0/4, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name Cayes
Switch(config-vlan)#vlan 99
Switch(config-vlan)#name DMZ
Switch(config-vlan)#interface range fa0/1 -10
Switch(config-if-range)#swichport access vlan 10
                        ^
% Invalid input detected at '^' marker.

Switch(config-if-range)#swichport access vlan 10
                        ^
% Invalid input detected at '^' marker.

Switch(config-if-range)#swichport mode access
                        ^
% Invalid input detected at '^' marker.

Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#interface fa0/24
Switch(config-if)#switchport access vlan 99
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG-I: Configured from console by console
```

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Configuration des commutateurs

 Switch2

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name Cayes
Switch(config-vlan)#vlan 99
Switch(config-vlan)#name DMZ
Switch(config-vlan)#interface range fa0/1 -10
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#interface fa0/24
Switch(config-if-range)#switchport access vlan 99
Switch(config-if-range)#exit
Switch(config)#exit
Switch#
Switch#en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname S3
S3(config)#interface range fa 0/1-24
S3(config-if-range)#switchport voice vlan 1
S3(config-if-range)#exit
S3(config)#
```

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Verification des address IP Automatique par le DHCP

The image shows a Cisco Packet Tracer network diagram and the configuration window for PC0. The network diagram on the left shows a central 2950 switch connected to three devices: PC-PT (PC0), IP Phone0, and IP Phone1. The switch is also connected to a 2611 router. The PC0 configuration window on the right shows the following settings:

IP Configuration	
Interface	FastEthernet0
IP Configuration	☒ DHCP ☐ Static DHCP request successful.
IPv4 Address	192.168.1.4
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	0.0.0.0

Below the IP Configuration section, the IPv6 Configuration section shows:

IPv6 Configuration	
IPv6 Configuration	☐ Automatic ☒ Static
IPv6 Address	
Link Local Address	FE80:2D0:BAFF:FEC7:9C77
Default Gateway	
DNS Server	

The 802.1X section shows:

802.1X	
Use 802.1X Security	☐
Authentication	MD5
Username	
Password	

The image shows a Cisco Packet Tracer network diagram and the configuration window for PC1. The network diagram on the left is identical to the one above, showing a central 2950 switch connected to three devices: PC-PT (PC0), IP Phone0, and IP Phone1. The switch is also connected to a 2611 router. The PC1 configuration window on the right shows the following settings:

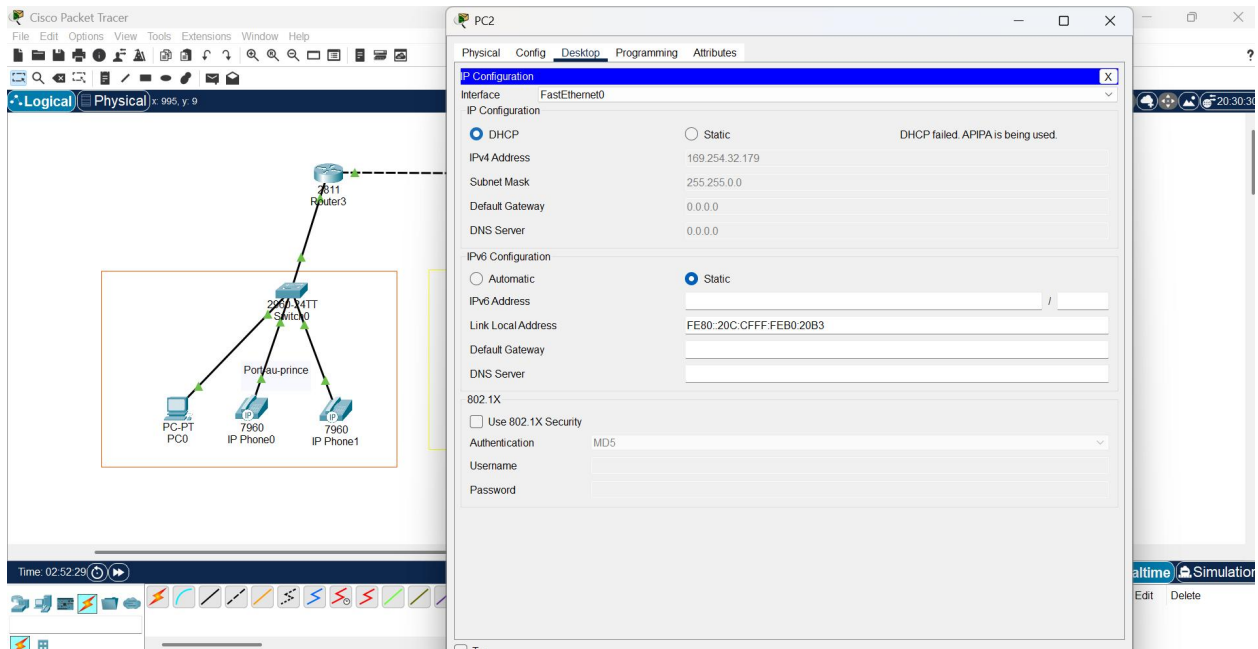
IP Configuration	
Interface	FastEthernet0
IP Configuration	☒ DHCP ☐ Static DHCP request successful.
IPv4 Address	192.168.2.4
Subnet Mask	255.255.255.0
Default Gateway	192.168.2.1
DNS Server	0.0.0.0

Below the IP Configuration section, the IPv6 Configuration section shows:

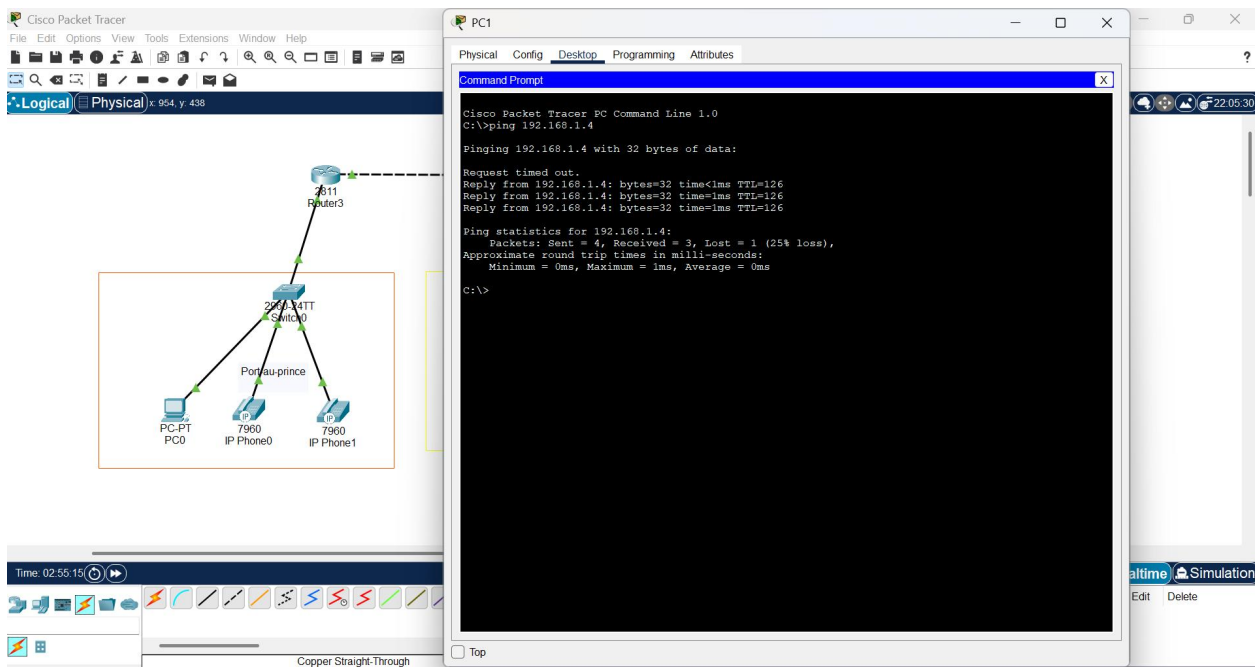
IPv6 Configuration	
IPv6 Configuration	☐ Automatic ☒ Static
IPv6 Address	
Link Local Address	FE80:203:E4FF:FEE6:441D
Default Gateway	
DNS Server	

The 802.1X section shows:

802.1X	
Use 802.1X Security	☐
Authentication	MD5
Username	
Password	



Test ping



Routage entre les deux routeurs (OSPF)

Premier router :

 Router3

Physical Config CLI Attributes

IOS Command Line Interface

```
Router con0 is now available

Press RETURN to get started.

%IPPHONE-6-REGISTER: ephone-1 IP:192.168.1.2 Socket:2 DeviceType:Phone has registered.
%IPPHONE-6-REGISTER: ephone-2 IP:192.168.1.3 Socket:2 DeviceType:Phone has registered.

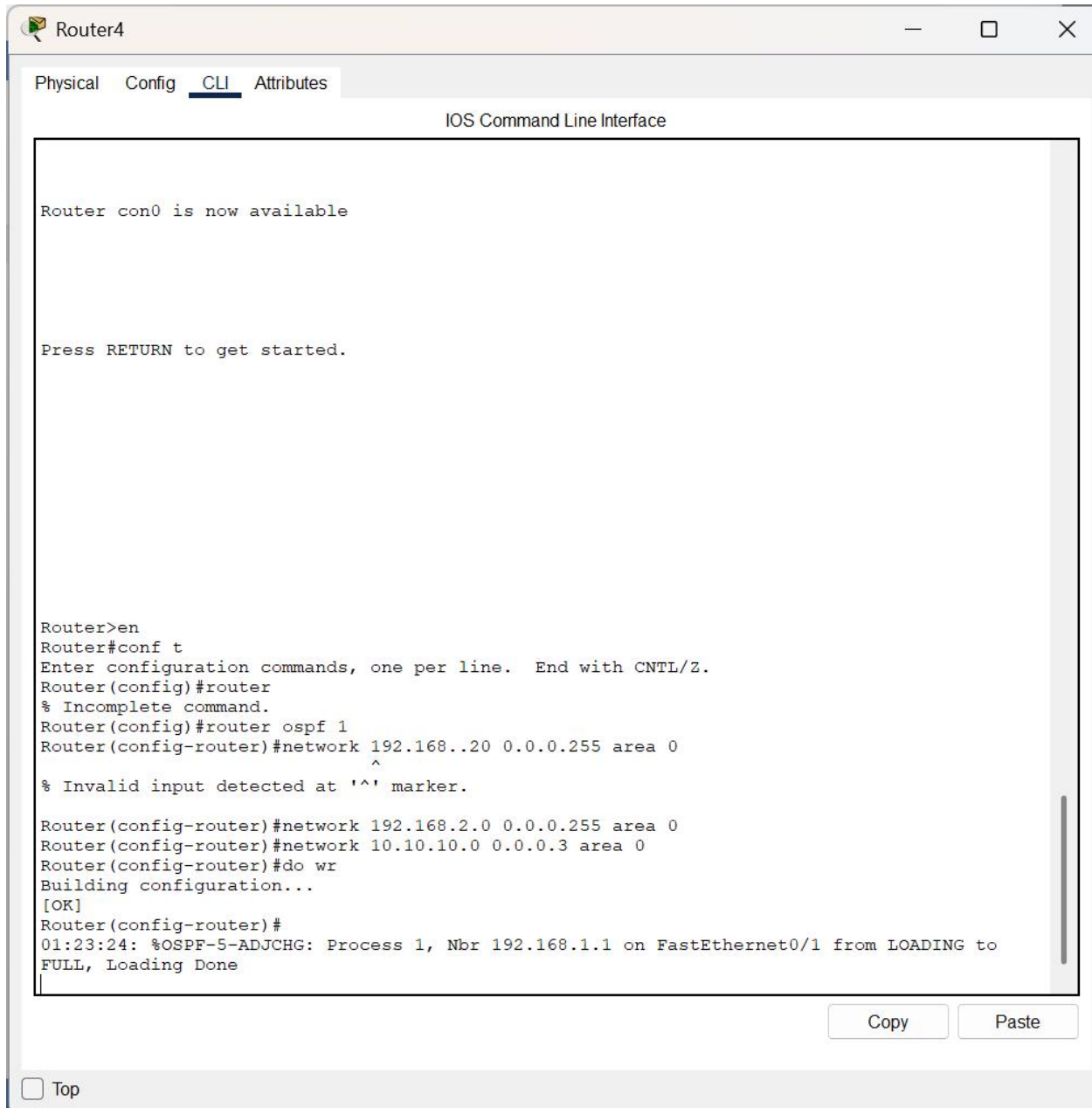
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ofps 1
      ^
% Invalid input detected at '^' marker.

Router(config)#router ospf 1
Router(config-router)#network 192.168.1.0 0.0.0.255 area 0
Router(config-router)#network 10.10.10.0 0.0.0.3 area 0
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#
```

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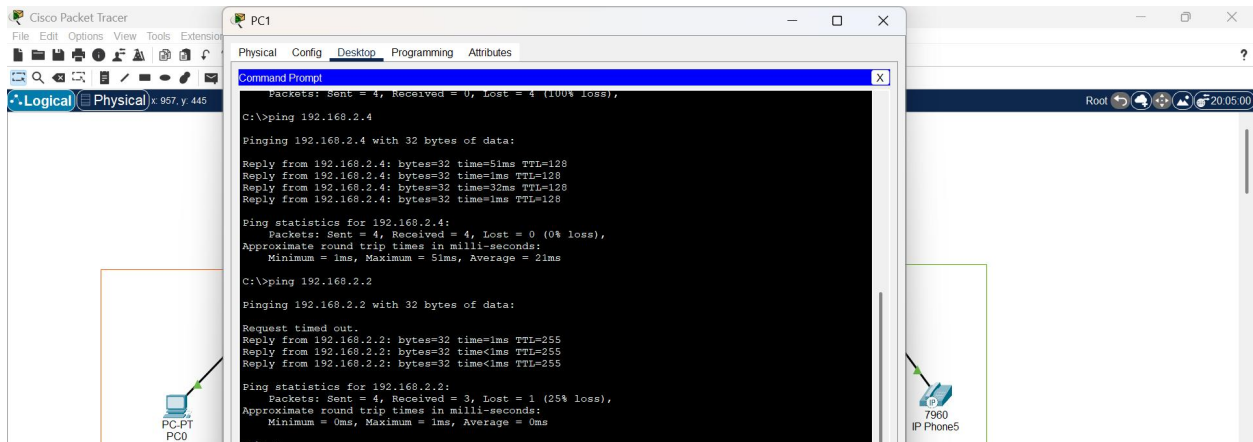
Deuxieme router :



Teste Pratique

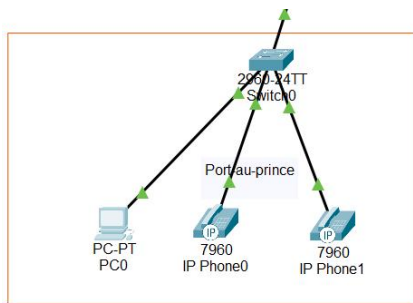
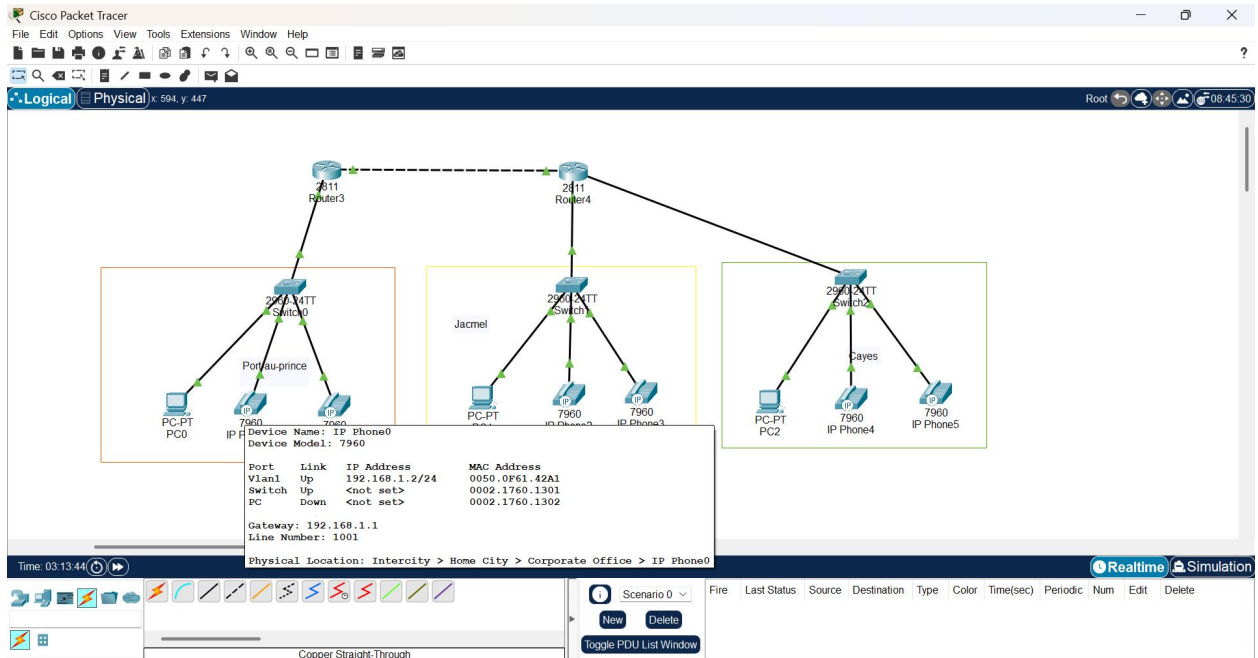
Test 1 : Connectivité inter-sites

ping Jacmel → Cayes



Test 2 : Attribution IP téléphones

DHCP OK



```

Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 53ms, Average = 16ms

C:\>ping 192.168.1.2

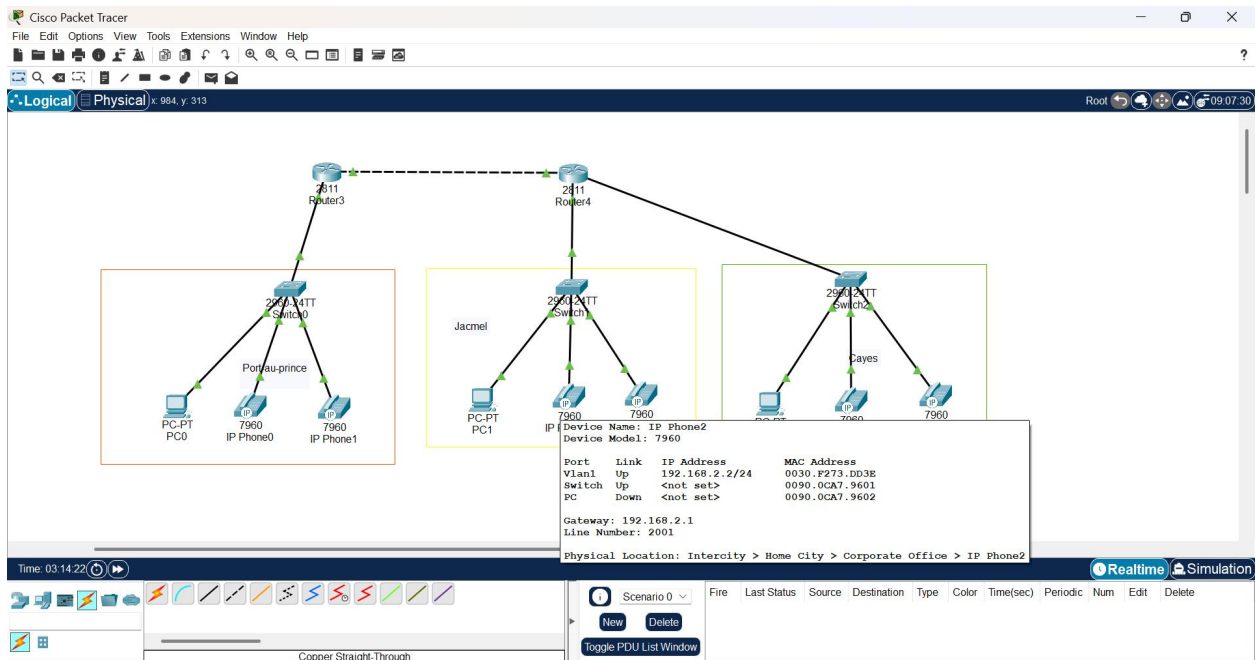
Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>

```



PC-PT PC0

```

Reply from 192.168.2.4: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.2.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 5ms, Average = 21ms
C:\>ping 192.168.2.2
Pinging 192.168.2.2 with 32 bytes of data:
Request timed out.
Reply from 192.168.2.2: bytes=32 time<1ms TTL=255
Reply from 192.168.2.2: bytes=32 time<1ms TTL=255
Reply from 192.168.2.2: bytes=32 time<1ms TTL=255
Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
  
```

7960 IP Phone5

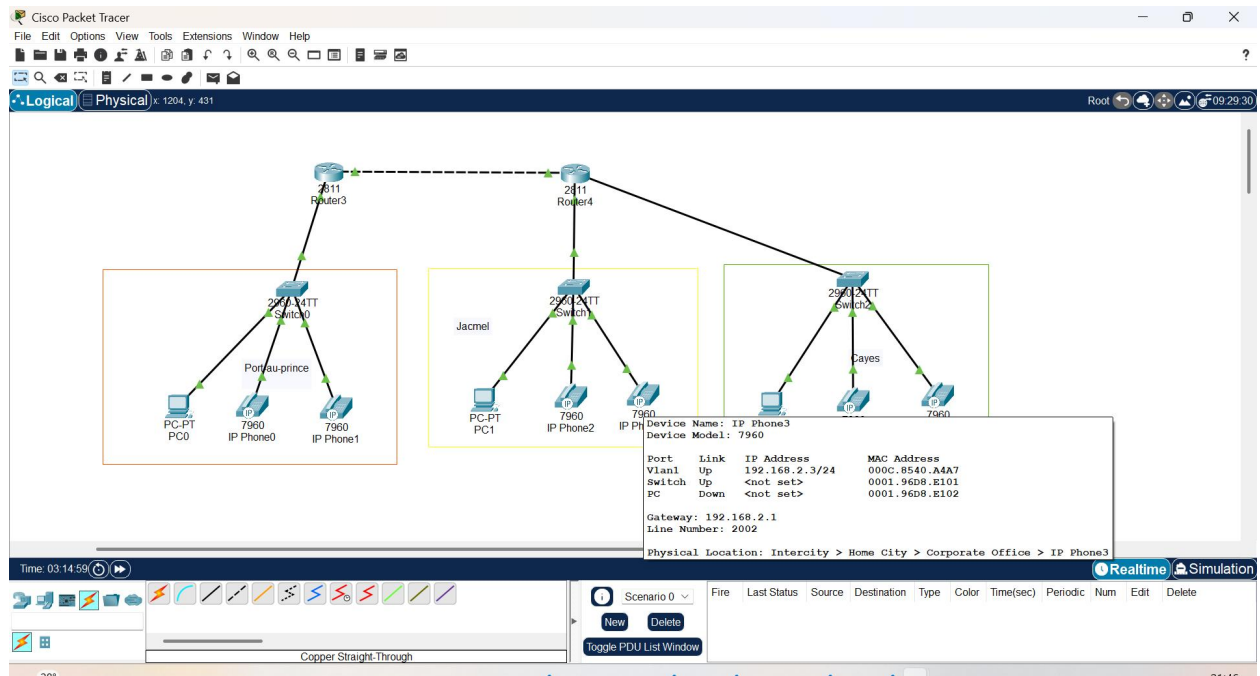
Time: 03:31:59

Realtime Simulation

Type Color Time(sec) Periodic Num Edit Delete

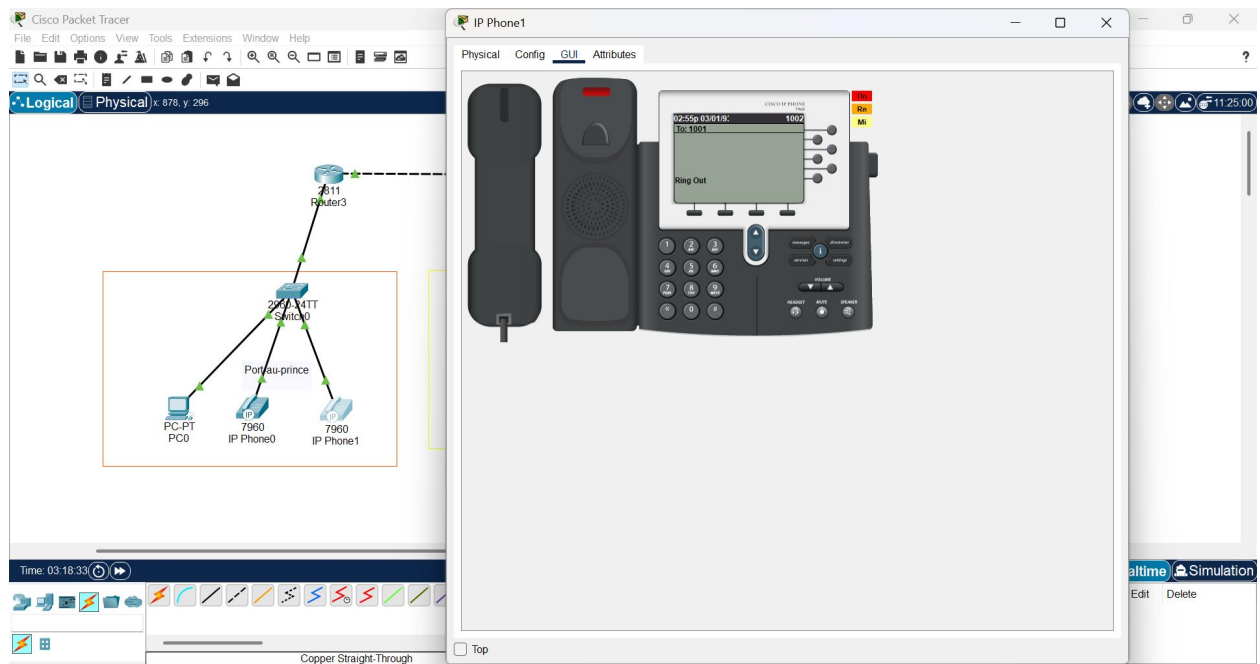
28°

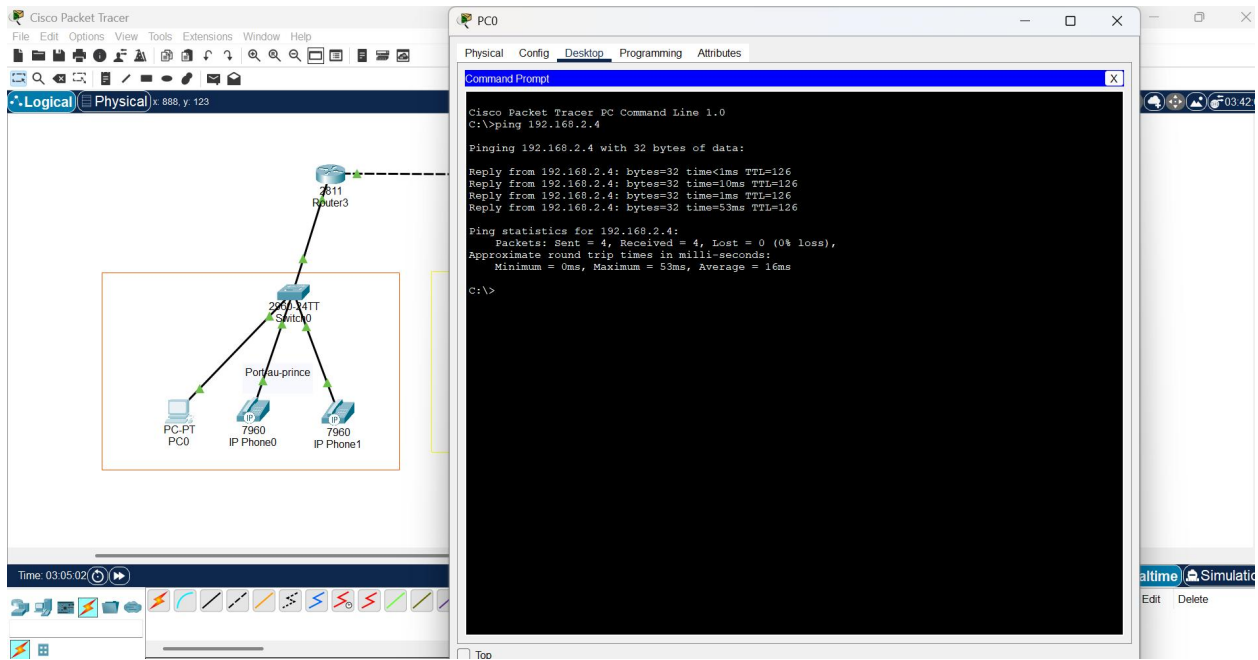
22:06



Test 3 : Appels VoIP

(P-au-P) → (Jacmel)





(Cayes) → (P-au-P) → (Jacmel)

